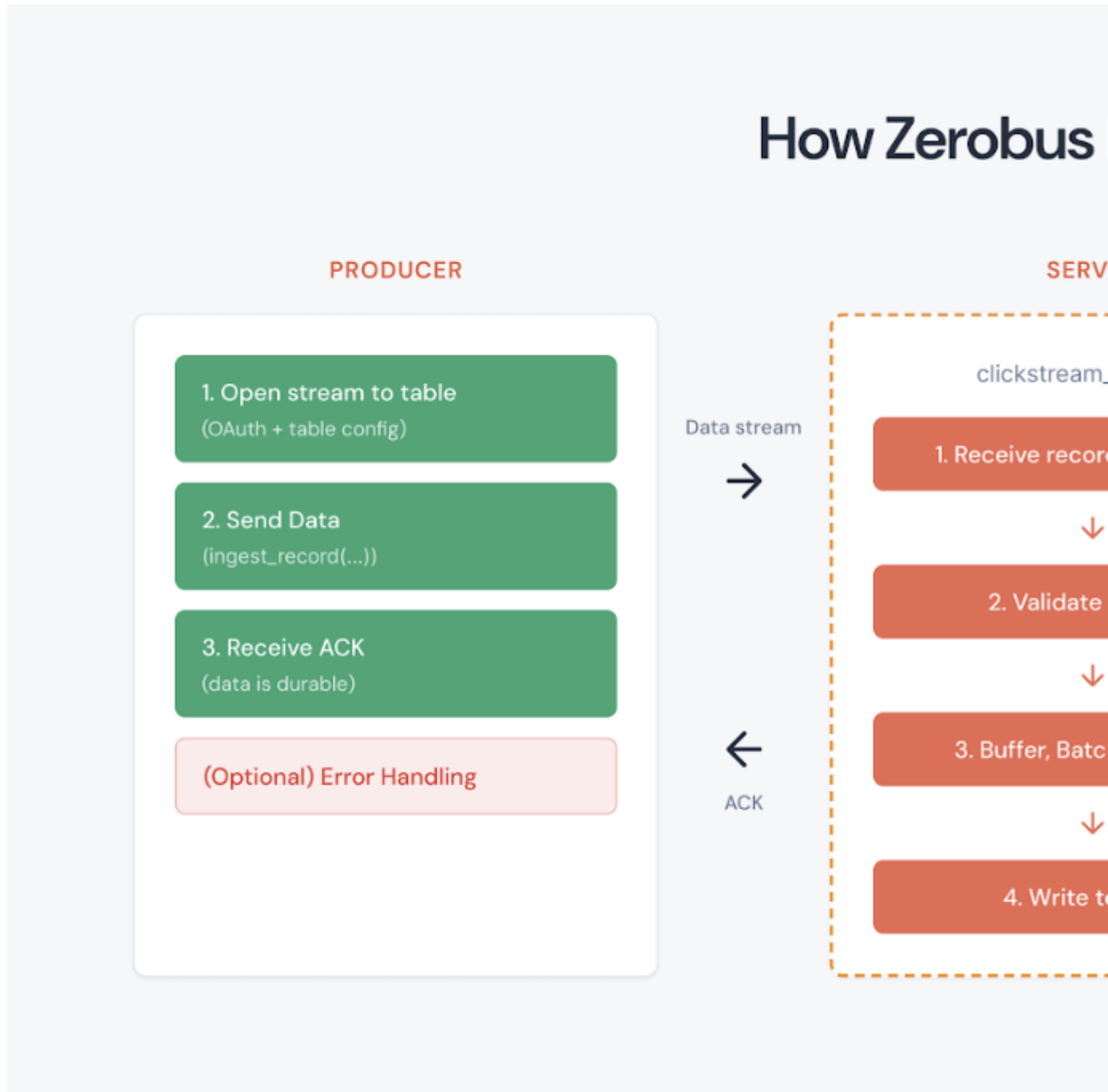


zerobus-feeder

Configurable Python client that streams synthetic data into a Databricks **Zerobus** endpoint.

Useful for load-testing, demos, and smoke-testing Zerobus integrations.



Features:

- CLI flags, YAML config file, or interactive wizard (your last entries are remembered and prefilled).

- Random-data generator driven by a simple JSON schema (strings, ints, longs, floats, doubles, booleans, timestamps, dates, binary, choice lists, nullable columns).
- Optional helpers that create the target Delta table and the service principal (plus OAuth secret and grants) via a Databricks CLI profile.
- Live terminal dashboard with throughput, latency percentiles, and a sparkline of recent latencies.
- Selectable transport: streaming **gRPC** (JSON or Protobuf), the **REST** API (Beta), **Arrow Flight** (Beta), or **OpenTelemetry / OTLP** (Beta) — see [Transports](#).
- OAuth2 client-credentials authentication — no PAT on disk.

Prerequisites

- Python 3.9+
- A Databricks workspace in a Zerobus-supported region (AWS, Azure, or GCP)
- Target table as a **managed** Delta table in Unity Catalog (the script can create it for you with `--create-table`)
- A service principal with OAuth2 client-credentials and `USE CATALOG` , `USE SCHEMA` , `MODIFY` , `SELECT` on the target table (the script can create and grant with `--create-sp / --create-table`)
- Databricks CLI configured (`~/ .databrickscfg`) if you want to use `--create-sp / --create-table` or auto-prefill `workspace_url / workspace_id`

Installation

Option A: startup scripts (recommended)

The `run.sh` (Linux / macOS) and `run.bat` (Windows) wrappers create a local virtualenv on first launch, install dependencies, and then forward all arguments to `zerobus_feeder.py` :

```
# Linux / macOS
git clone https://github.com/guidooswaldDB/zerobus-feeder.git
cd zerobus-feeder
./run.sh
```

```
REM Windows
git clone https://github.com/guidooswaldDB/zerobus-feeder.git
cd zerobus-feeder
run.bat
```

Subsequent runs reuse the existing venv and skip reinstalling unless `requirements.txt` has changed.

Option B: manual

```
git clone https://github.com/guidooswaldDB/zerobus-feeder.git
cd zerobus-feeder
python3 -m venv .venv
source .venv/bin/activate          # Windows: .venv\Scripts\activate
pip install -r requirements.txt
```

Quick start (first run)

On the first invocation there is no remembered state, so the script offers a guided setup:

```
./run.sh          # or: run.bat    - or: python zerobus_feeder.py
```

The wizard will:

1. Let you pick a Databricks CLI profile from `~/ .databrickscfg` and prefill `workspace_url` / `workspace_id` from it.
2. Optionally create a new service principal and OAuth secret.
3. Optionally create the target table (from the schema JSON) and apply grants.
4. Prompt for any remaining required fields and start streaming.

After the first run, values are written to `.zerobus_feeder_last.yaml` (already listed in `.gitignore`) and prefilled next time.

Configuration precedence

Source	Notes
<code>--config <yaml></code>	If passed, all other CLI flags and remembered values are ignored.
CLI flags	Override remembered values.

Source	Notes
<code>.zerobus_feeder_last.yaml</code>	Remembered values from the previous run.
Interactive prompts	Fill anything still missing (unless <code>--non-interactive</code>).

CLI reference

```
python zerobus_feeder.py [options]
```

```

--config, -c FILE          Path to YAML config file (overrides all other params)
--eps FLOAT                Target events (records) per second
--schema-file FILE         JSON file describing the data structure
--workspace-id DIGITS      Used to build the Zerobus endpoint
--region CODE              e.g. us-west-2, eastus
--cloud {aws,azure,gcp}    Cloud provider
--table-name NAME          Full table name (catalog.schema.table)
--transport {grpc,rest,arrow}
                           How to reach Zerobus (default grpc). See Transports.
--record-format {json,protobuf}
                           gRPC serialization (default json); ignored by rest/arrow
--client-id ID             Service principal client ID
--client-secret SECRET     Service principal client secret
--workspace-url URL        https://<host>; auto-filled from CLI profile if set
--profile NAME             Databricks CLI profile (for --create-sp/--create-table)
--warehouse-id ID         SQL warehouse used by --create-table
--sp-display-name NAME     Display name for --create-sp
--create-sp                Create an SP + OAuth secret (needs --profile)
--create-table             Create the target table from the schema (needs --profile)
--no-table-latency         Skip the periodic ingest-latency SQL query
--interactive              Force prompts for every parameter
--non-interactive           Never prompt; error on any missing required parameter

```

Mandatory parameters: `eps`, `schema_file`, `workspace_id`, `region`, `cloud`, `table_name`, `client_id`, `client_secret`, `workspace_url`. In `--non-interactive` mode the script exits with code 2 and lists any missing ones.

YAML config

`sample_config.yaml` documents every parameter. When passed via `--config`, it is the sole source of configuration:

```
python zerobus_feeder.py --config sample_config.yaml
```

Transports

The feeder can reach Zerobus four different ways, selected with `--transport` (or `transport:` in YAML). The probe/latency cycle and dashboard work across all of them — only the wire path changes, which makes this useful for comparing them head-to-head. For Databricks' own guidance on picking an interface, see [Choose an interface](#).

Transport	Wire	Serialization	Status	Needs
grpc (default)	SDK streaming gRPC	JSON or Protobuf	GA	databricks-zerobus-ingest-sdk
rest	HTTPS POST	JSON	Beta	requests (no SDK)
arrow	Arrow Flight (gRPC)	Apache Arrow RecordBatch	Beta	...-sdk[arrow]>=1.3.0 + pyarrow
otlp	OTLP/gRPC	OpenTelemetry spans / logs / metrics	Beta	opentelemetry-sdk + OTLP gRPC exporter

```
# gRPC with Protobuf (descriptor built from the schema at runtime)
python zerobus_feeder.py --transport grpc --record-format protobuf

# Zerobus REST API (no SDK needed)
python zerobus_feeder.py --transport rest

# Arrow Flight
python zerobus_feeder.py --transport arrow

# OpenTelemetry (OTLP) - table-name is the OTel table PREFIX
python zerobus_feeder.py --transport otlp --create-table \
  --table-name catalog.schema.myprefix --profile my-profile
python zerobus_feeder.py --transport otlp --table-name catalog.schema.myprefix
```

Notes:

- **gRPC** is the original path. `--record-format protobuf` compiles a protobuf message from `schema_file` in-process (no `.proto` file or `protoc` step) and streams binary records; `json` (default) streams JSON.
- **REST** authenticates with the OAuth2 client-credentials flow against `<workspace_url>/oidc/v1/token` and POSTs a one-element JSON array to `https://<endpoint>/zerobus/v1/tables/<table>/insert`. Each POST is a

synchronous, durable write — there is no fire-and-forget path — so the "streaming" phase is round-trip-bound and high `--eps` targets may not be reached. That's an honest characteristic of the REST API, not a bug.

- **Arrow** (Beta) sends one-row `pyarrow.RecordBatch` objects over Arrow Flight. It requires `databricks-zerobus-ingest-sdk[arrow]>=1.3.0`; `pip install -r requirements.txt` installs the right version. Delta `TIMESTAMP` maps to `timestamp('us', tz='UTC')` and `DATE` to `date32`.
- **OTLP** (Beta) sends synthetic OpenTelemetry **spans, logs, and metrics** over OTLP/gRPC, rotating across the three signals so all three OTel tables are exercised. This is the odd one out:
 - `--table-name` is a **prefix** (`catalog.schema.prefix`); the feeder writes to `<prefix>_otel_spans`, `<prefix>_otel_logs`, and `<prefix>_otel_metrics`.
 - The data-structure JSON does **not** apply — OTLP data follows the fixed OTel v2 schema. `--create-table` creates the three predefined tables (with `CLUSTER BY (time, service_name)` and `otel.schemaVersion=v2`) and grants. If you start an OTLP run and the tables don't exist yet, the feeder detects that (when a `--profile` is set) and offers to create them on the spot.
 - Each signal table gets its own OAuth token (per-table `authorization_details`) and is addressed with the `x-databricks-zerobus-table-name` header.
 - Authentication is the same client-credentials flow as REST. The SQL table-latency panel is auto-disabled (the OTel schema has no `event_time`).
 - Like REST, each export is a synchronous round-trip, so high `--eps` targets are round-trip-bound. For production OpenTelemetry you'd typically point an **OpenTelemetry Collector** at Zerobus and let it handle token refresh; this feeder talks to the endpoint directly for latency comparison.

Protobuf and Arrow derive their schema from the same data-structure JSON, so the target table must match it (use `--create-table` to keep them in sync). OTLP uses its own fixed schema instead.

Data structure JSON

`sample_schema.json` shows the supported types and per-column options. The schema drives both the randomly generated payloads and the `CREATE TABLE` issued by `--create-table`.

Type	Delta type	Options
string	STRING	length or min_length / max_length , prefix , alphabet , choices

Type	Delta type	Options
int	INT	min , max , choices
long	BIGINT	min , max , choices
float	FLOAT	min , max , precision
double	DOUBLE	min , max , precision
boolean	BOOLEAN	true_probability
timestamp	TIMESTAMP	offset_seconds , jitter_seconds (emitted as Unix microseconds)
date	DATE	offset_days , range_days (emitted as epoch days)
binary	BINARY	length (base64-encoded in JSON)

Every column also accepts `nullable: true` + `null_probability: 0.0..1.0`.

Creating the service principal and table

```
python zerobus_feeder.py \
  --profile DEFAULT \
  --schema-file sample_schema.json \
  --table-name main.default.zerobus_feeder_events \
  --cloud aws --region us-west-2 --workspace-id 1234567890123456 \
  --create-sp --create-table --eps 50
```

This will:

1. Create a service principal named `zerobus-feeder` (override with `--sp-display-name`), generate an OAuth secret, and store the credentials in `.zerobus_feeder_last.yaml`.
2. Prompt for (or auto-pick) a SQL warehouse, issue `CREATE TABLE IF NOT EXISTS <table>`, then grant `USE CATALOG` / `USE SCHEMA` / `MODIFY` / `SELECT` to the SP.
3. Start the stream at 50 EPS with the live dashboard.

If the workspace-level OAuth secret API isn't available, the script prints a link to the Databricks UI so you can generate the secret manually, then rerun with `--client-id` / `--client-secret`.

Non-interactive / automated runs

After a first successful run, you can reproduce the stream from remembered state:

```
python zerobus_feeder.py --non-interactive
```

Or from a YAML file (ideal for CI / ephemeral VMs):

```
python zerobus_feeder.py --config /path/to/config.yaml
```

Live dashboard

While running, the terminal shows:

- Target vs. actual EPS (events per second), total sent, errors, elapsed time
- Current / min / max / p50 / p95 / p99 SDK round-trip latency (ms)
- Unicode-block sparkline of recent per-record latencies
- End-to-end ingest latency (event_time → file_modification_time) over the last 15 minutes, sampled via a SQL query against the target table after every probe batch — disable with `--no-table-latency`
- Target table, Zerobus endpoint, workspace URL


```
guido. ) ▶ ~/code/zerobus-f
Connecting endpoint=739.zerobus
Workspace URL https://adb-739.1
Table guido.zerobus.feed2
Authenticating as client_id=5f2aa102-67cf-4
✓ Stream opened. Starting ingestion...
```

Status:	• quer
Sent:	6,900
Elapsed:	0:17:3
Latency now:	200.90
Latency min / max:	121.00
Target:	guido
Table latency (15m):	avg 2

Probe

321.5 ms

249.2 ms

Press `Ctrl+C` to flush and close the stream cleanly; a final summary is printed.

Ingest-latency query

When a `profile` and `warehouse_id` are configured (or `warehouse_id` is selected at startup when missing), the feeder executes this query once per probe cycle on the same SQL warehouse:

```
SELECT
  count(*)
  round(avg(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(event_time))
  min(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(event_time))
  max(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(event_time))
  percentile_approx(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(event_time), 0.5)
  percentile_approx(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(event_time), 0.9)
  percentile_approx(unix_timestamp(_metadata.file_modification_time) - unix_timestamp(event_time), 0.99)
FROM <catalog>.<schema>.<table>
WHERE event_time >= current_timestamp() - INTERVAL 15 MINUTES;
```

The query is bounded by a 15s warehouse-side timeout, so a cold warehouse cannot stall streaming. Pass `--no-table-latency` to skip it entirely.

Authentication

Zerobus is authenticated via **OAuth2 client-credentials (M2M)** using a Databricks service principal. `client_id` and `client_secret` are passed to the Zerobus SDK, which handles token acquisition and refresh. The Databricks CLI profile is only consulted for the optional `--create-sp` / `--create-table` helpers and to prefill `workspace_url` / `workspace_id`.

Files

File	Purpose
<code>zerobus_feeder.py</code>	The script
<code>run.sh</code>	Startup wrapper for Linux / macOS (creates venv, installs deps, runs the script)
<code>run.bat</code>	Startup wrapper for Windows (same behaviour as <code>run.sh</code>)
<code>sample_schema.json</code>	Example data-structure definition

File	Purpose
<code>sample_config.yaml</code>	Example YAML config
<code>requirements.txt</code>	Python dependencies
<code>.gitignore</code>	Excludes the last-values file, log file, venvs, caches
<code>.zerobus_feeder_last.yaml</code>	Auto-generated, local-only, stores last-used values including the client secret — never commit this
<code>zerobus_feeder.log</code>	Auto-generated, local-only, detailed per-session log (config resolution, SP/table creation, DDL, periodic latency snapshots, errors). Appended across runs. Never commit this

Troubleshooting

- **Authentication failed / error 4024**: the SP is missing explicit table-level `MODIFY / SELECT`. Schema-level inherited grants aren't sufficient for the Zerobus `authorization_details` flow.
- **Connection refused / DNS errors**: verify `cloud` and `region` match the workspace; the endpoint is built as `<workspace_id>.zerobus.<region>.<cloud-suffix>`.
- **Actual EPS well below target**: latency dominates at the single-stream throughput limit (100 MB/s, 15 k rows/s). Run multiple instances or lower the per-record payload.
- **First-run wizard doesn't appear**: it triggers only when `.zerobus_feeder_last.yaml` is absent and `--config / --non-interactive` aren't set. Delete the file to reset.

Coming soon

- OTEL (Open Telemetry) – OTLP/gRPC
- gRPC (Arrow)

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